



Group 17319

Capstone Portfolio

**Solving Traffic Congestion Using AI-based
traffic lights management and
V2V Communication**

Grade 12 – (2025/2026) – Semester 1

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Introduction

Traffic in Egypt has honestly become something we all just live with, even though no one actually likes dealing with it. If you've ever tried getting across Cairo during the day, you already know how it feels: the cars barely move, people get impatient, and you basically watch your time disappear for no good reason. It's not only about being late — traffic burns extra fuel, fills the air with smoke, and makes the whole city feel more stressful than it should be. While working on this project, we kept asking ourselves why the situation feels like it never changes. And the thing is, the traffic system itself doesn't really "react." The lights switch at the same time whether there are five cars or fifty. Emergency vehicles get stuck behind the same lines as everyone else. And a single driver pressing the brakes suddenly can slow down an entire street for minutes. It's like the roads are running on old habits instead of actual information. So our idea was to try something different: a traffic model that actually pays attention. Instead of letting cars pile up randomly, we wanted the system to notice what's happening and respond instantly. We followed the Engineering Design Process, step by step, to build a small version of a smart intersection. It uses a camera to count cars, sensors to measure distances, and a setup that changes the timing of the traffic lights based on the situation instead of guessing. We added something for emergency vehicles too — a green corridor that clears a path when needed, because an ambulance shouldn't be stuck behind someone who's checking their phone. And since sudden stops create half the traffic in Cairo, we tried to keep a steady distance between cars in the model so everything moves smoothly without those chain reactions that usually block the road. At the end of the day, this project isn't just about electronics and sensors. It's about showing that Cairo doesn't have to be impossible to move through. If the system becomes smarter, people waste less fuel, breathe cleaner air, and simply get their time back. Even a small improvement would make a noticeable difference in daily life.

Chapter 1 | Present and Justify Solution Requirements

Egypt Grand Challenges

For the last few years, Egypt faced many challenges that became obstacles to its development, and the government has tried hard to solve these problems to continue on its road of success and improvements in many fields.

Defining these problems is necessary to solve them, as there are too many challenges. The economic researchers chose 12 challenges to be the main grand challenges in Egypt and to focus on solving them, as they include the most important fields in Egypt.

These 12 main grand challenges are shown below:

- ❖ *Deal with exponential population growth.*
- ❖ *Reduce pollution.*
- ❖ *Work to eradicate public health issues.*
- ❖ *Improve the use of alternative energies.*
- ❖ *Reduce urban congestion.*
- ❖ *Reduce and adapt the effects of climate change.*
- ❖ *Improve the use of arid areas.*
- ❖ *Increase industrial base for Egypt.*
- ❖ *Improve sources of clean water.*
- ❖ *Increase opportunities for Egyptians to stay and work in Egypt.*
- ❖ *Recycle and retain garbage for recycling.*
- ❖ *Improve the scientific and technological environment for all.*

Deal with population growth and its consequences

Population growth is a demographic concept that shows the increase in the population of a country. It can be represented by subtracting the number of mortalities from the number of births per year. Having a high population growth is not considered a problem, the problem comes when there is a lack of resources in the country and the population is getting higher, so there will not be enough resources for the new generations which leads to many other problems such as poverty, bad education, Unemployment, urban congestion and in some cases, it could lead to famines.

For Egypt, over the last two decades, the country has been through rapid population growth which put a large pressure on the economy and the projects of enhancing the life quality. The population growth ratio in Egypt in 2023 was 1.5. as shown in Graph (1) and the population was about 111 million people according to the World Bank.



Graph 1: (Shows the population growth in Egypt)

This increase in the population is caused by many reasons:

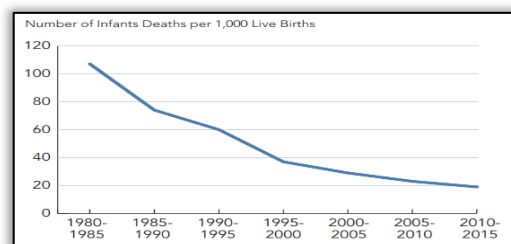
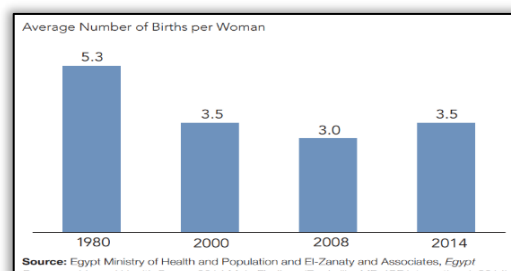
- High fertility rate.
- Low mortality rate.
- Young girls' marriage.

Egypt's population more than tripled in the second half of the 20th century, this is because of the increase in births number according to the high fertility rate (Average Number of Births per Woman), and the decline in mortality number especially in children. *As shown in Graphs (2), (3).*

Children born in Egypt today are expected to live 71 years on average, 10 years longer than those born around 1985. This is mainly because of the improvements in health care in the last few years. allowing a much larger percentage of newborns to reach adulthood and in turn have their own children. Today, Egypt's infant mortality rate is about half of the world's average.

Marriage is nearly universal in Egypt and many families still marry off their daughters at a young age. In 2014, more than an half a million girls ages 15 to 19 were ever married.⁷ For these girls, early marriage generally means early childbearing, adding momentum to the country's population growth.

Thus, the country is facing many challenges to improve the life quality of the people, every 100 people responsible for production in Egypt work to provide the needs of 60 other people, which works to reduce average incomes as well as averages of savings and investment at the state level. The demand for food has become greater than the supply, which led to the import of a large volume of basic food resources such as wheat and meat. The Egyptian state has spent in the last ten years 9 times what was spent in 2014/2015, in the field of education to create new classrooms simultaneously. The Egyptian state spent this year 13 times what was spent on investment in health 10 years ago.



Graphs 2 & 3: (show fertility and mortality rates in Egypt)

Adress and reduce pollution fouling air, water, and soil.

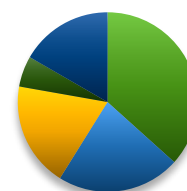
One of the most important problems that the world faces is pollution. Increasing pollution in Egypt risks a lot of people's lives. There are many types of pollution such as air pollution and water pollution. The most common type of pollution that Egypt suffers from is air pollution.

Air pollution

This is the main type to put a spot on to gain more information and understand the challenge this semester.

Air pollution is primarily caused by carbon and nitrogen-based compounds. Carbon, mainly in the form of carbon dioxide (CO₂) and carbon monoxide (CO), is produced by burning fossil fuels. Nitrogen oxides (NO_x), released from vehicles and industrial activities, contribute to smog and acid rain. Both carbon and nitrogen compounds have significant environmental and health impacts. Air pollution has been a problem for decades in Egypt. Air pollution was the cause of 90.559 premature deaths in Egypt in 2019, which is more than road injuries and diabetes. The current air quality in big cities in Egypt like Cairo is poor, and that is because of bad human actions and habits.

Causes of air pollution in Egypt



Graph 3

- Transport
- Burning wastes from agriculture
- Industry
- mismanagement of wastes
- others

Graph 4: Causes of air pollution in Egypt.

The causes of air pollution in Egypt:

- 1)Transport
- 2)Burning wastes from agriculture
- 3)Industry
- 4)mismanagement wastes



Figure 1: air pollution in Egypt.

As shown in Graph (3).

Many other resources pollute the air of Egypt.

The first cause is (transport). The traffic in Cairo has reached 80% of the congestion problem. Traffic volume can reach 7,000 vehicles per hour per lane, which makes a huge amount of greenhouse gases.

Burning agricultural waste has become more common because of the increase in agricultural areas near the valley of the Nile. In 2020, the government collected 500,000 tons of straw, which is 90% of the production.

Industrial power is the third-highest concentration of pollution in Cairo. The factories' lands, which are separated in many different places, cause air pollution by the gases and smoke that they produce while working. Most of these industries are naturally based, but the way they are used harms the air.

In 2023 Egypt has been ranked globally in air pollution as the 9th in the world which is proof of the huge problem for the sensitive group of people in this country.

Water pollution

The Nile River is the main source of water in Egypt. The river is continuously contaminated



Figure 2: water pollution in Egypt.

with overwhelming amounts of items such as discharge, toxic chemicals, fertilizer residue, radioactive waste, and oil pollution is truly horrific and dangerously deadly. Egyptian traditions can be considered a large cause of water pollution.

The pollution of the Nile is one of the most serious issues that faces Egypt.

About 4BCM of agriculture and industrial wastes containing acids, organic materials, and heavy metals are discharged into the Nile and its branches. Iron and manganese concentrations in the delta reach 1,600 µg/liter and 220 µg/liter, respectively. The activities of polluted water in the delta have led to an increase in the concentration of chemical nutrients such as nitrogen.

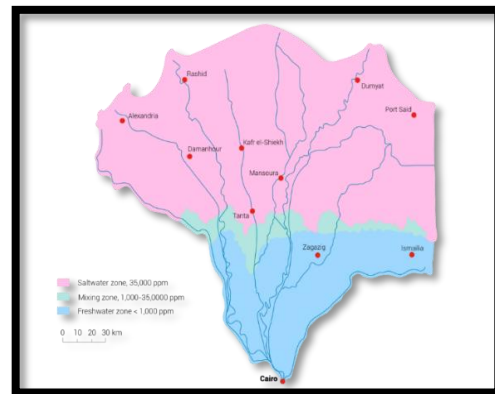


Figure 3: The increasing of swamps in the Delta

Climate change and sea level rise made a problem, which is the seawater infiltration to the Delta region by allowing vast quantities of saltwater to penetrate the depths of the adjacent aquifers. The groundwater in the Nile Valley is often of higher quality than that in the Nile Delta. At a depth of 300 meters, seawater has surged to Mansoura, which is approximately 70 km from the coast.

Soil pollution

Soil pollution in Egypt has severe environmental and health consequences, particularly in agricultural regions like the Nile Delta. Key pollutants include heavy metals, such as lead and cadmium, which accumulate due to industrial waste, mining, and improper disposal of hazardous materials. These toxins bind to soil particles, impairing soil structure and reducing its fertility. Excessive use of chemical fertilizers and pesticides introduces nitrates and phosphates, disrupting nutrient cycles and leading to eutrophication in nearby water bodies. This pollution affects microbial biodiversity, reducing beneficial organisms vital for nutrient decomposition and soil health. Long-term exposure to contaminated soil also increases the risk of chronic diseases, including cancer, due to bioaccumulation of toxins in crops and groundwater. Addressing these issues in Egypt requires advanced soil remediation techniques, like phytoremediation, and adopting integrated pest management (IPM) for sustainable agriculture.

There are many types of pollution that Egypt suffers from such as (Noise pollution – Waste pollution – Light pollution). All these types don't cause the same harm as Air and water pollution do, especially in a country with this number of populations like Egypt.

We will try to help the world to reduce pollution.

Work to eradicate public health issues and diseases

Egypt's new Constitution 2014 powerfully addresses health as a fundamental human right and declares commitment to covering the whole population with quality health services. Egypt's progress towards universal health coverage (UHC) has been spurred recently by approving the Social Health Insurance (SHI) law, ensuring adequate and sustainable funding for health and reducing out-of-pocket expenditures. The new law has been approved by the government and parliament and authorized by the president for initiating implementation.

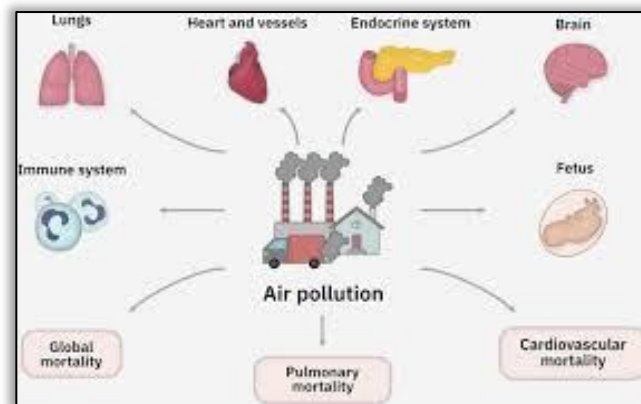


Figure 4: (shows the effect of air pollution on human health.)

Egypt has a long experience developing and implementing health system reforms and engaging with development partners. It has an elaborate network whereby 95% of the population lives within a 5 km radius of a given health facility. Commitment to the family practice program has expanded to cover 50% of primary care facilities and developed a national plan for scaling up family practice. The health information system within MoHP handles a multitude of indicators generated by the health sector with complete civil registration and vital statistics (CRVS) system. Sustained health system support has helped achieve high immunization coverage, Schistosomiasis control, and elimination of polio, diphtheria, pertussis, and lymphatic filariasis, besides improvement towards the achievement of MDG goals. Noncommunicable diseases (NCDs) constitute a significant burden on the health system in Egypt and are considered a substantial challenge for socioeconomic development in the country. About 84% of total mortality

in Egypt is attributed to NCDs, mainly Cardiovascular diseases, Cancer, Chronic Lung diseases, and Diabetes. Also, the prevalence of main behavioral and biological risk factors for these diseases is very high (Tobacco use, physical inactivity, unhealthy diet, obesity, hypertension, and hypercholesteremia).

Microplastics entering the human body via direct exposures through ingestion or inhalation can lead to an array of health impacts, including inflammation, genotoxicity, oxidative stress, apoptosis, and necrosis, which are linked to a variety of adverse health outcomes including cancer, cardiovascular diseases, and a lot of other conditions also other negative health issues that the plastic reasoned in are Direct toxicity as in the cases of lead, cadmium, and mercury

Also, Carcinogens, as in the case of diethylhexyl phthalate (DEHP), so in a way to decrease these problems, recycling the plastic polymers and taking them from the environment to insert them for improving particular industries will reduce the health issues and diseases caused by this plastic lead to a healthier hygienic environment.

Cutting down on the sources of pollution is important. Governments should make stricter rules for factories, vehicles, or any machine that increases air pollution, the government should encourage cleaner options like electric cars and renewable energy, such as solar and wind power. Cities can be designed with more green spaces and better public transportation to reduce pollution, and monitoring air quality in real-time can help people make better decisions for their health.

We also need more research to develop technology that reduces pollution and to find better treatments for diseases caused by it with stronger policies, cleaner energy, and public awareness, we can reduce the harmful effects of air pollution and make communities healthier.

Deal with urban congestion

Urban congestion in Egypt is a major problem caused by overpopulation, insufficient infrastructure, and increased car ownership. It manifests in traffic jams, higher air pollution, and a reduced quality of life. The government is taking steps to address it, but more needs to be done to improve the situation.

Egypt is one of the most crowded countries, the number of people in it is increasing a year after another especially in its capital Cairo, which is known as the Greater Cairo Metropolitan Area (GCMA) (**As shown in figure 4**)

Cairo is the most crowded place in Egypt. It contains more than 19

Figure 3

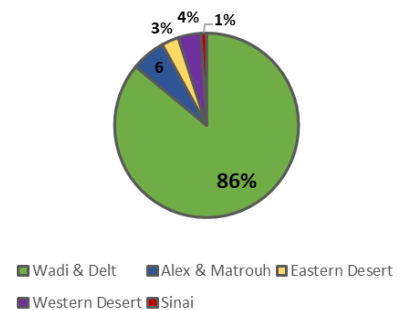
million inhabitants and the scientist expect an increase in this number by 2027 to reach 24 million (**As shown in figure 3**). Cairo contains the fifth of the population of Egypt.

Despite Egypt's large area, the population is concentrated in Cairo, which its area is 3,085 km².

The area of Egypt is 1.002 million km², which is a very huge space that Can handle a lot of residents, but the most people are concentrated in Cairo and in Nile River Delta. Number of births in Egypt in 2020 is 2,193,000 people.

This huge Urban congestion leads to lots of problems as the huge traffic (**As shown in figure 3**) in the streets which leads to:

1- Economic costs: Traffic congestion costs Egypt an estimated EGP 50 billion (US\$8 billion) per year, or about 4% of its GDP. This



Graph 5: The number of people living in certain areas in Egypt

is due to wasted fuel, lost productivity, and increased travel times.

2- Environmental costs: Traffic congestion is a major source of air pollution in Egypt. Because of the harmful gases emitted from the cars and factories. This can lead to respiratory problems, heart disease, and other health problems.

3- Social costs: Traffic congestion can make it difficult and stressful to get around, which can reduce people's quality of life. It can also lead to social isolation and increased crime rates.

All that leads to the waste of time and burn of more fuel which increases the air pollution and increases the price of transportation and causes lots of other problems as increasing the risk of accidents and traffic-related injuries.

So, we have to do lots of things to reduce these harmful effects as:

1- Investing in public transportation: Public transportation is a more efficient and environmentally friendly way to travel than private cars. The Egyptian government should invest in expanding and improving public transportation services, such as the metro and bus systems.

2- Discouraging car ownership: The Egyptian government can discourage car ownership by making it more expensive to own and operate a car. This can be done through taxes, parking fees, and other measures.

By that, we can reduce the effect of Urban congestion, but to stop it, we have to find the core of the problem and take some steps to end it.

There are many reasons why people prefer staying in Cairo over any other place in Egypt, but I want to focus specifically on one of the most important points from my perspective.

It's about the Availability of services, especially the availability of fresh drinking water, but now these days Egypt suffers from a shortage of water due to the big

increase of number of people and their huge need of water so we should all think with a creative method for finding a solution for this huge problem.

Problem to be solved

Traffic congestion and its consequences in highly concentrated squares in Egypt

The Greater Cairo Region (GCR) is known to be represent about 19% of Egypt's population, as it had about 16.1 million inhabitants in 2018, and this number has grown in the past few years. Creating all kinds of urban problems, such as urban congestion and urban traffic. This leads to other consequences like air pollution, health issues, etc. The primary source of PM2.5 air pollution in the greater Cairo area is road transport (33%), with ageing private cars and taxis accounting for 80% of the extreme congestion problem. Traffic volumes can reach 7,000 vehicles per hour per lane, far exceeding the maximum theoretical flow of 1,900.

This huge number of vehicles in this area isn't even organized. In 2024, Egypt recorded 5,260 road accident fatalities. Cairo had the highest number of accidents with 766 accidents in 2024.

The traffic system in Egypt is nearly to verge of a disaster. Cairo's seemingly never-ending traffic costs Egyptians EGP 47 billion in losses every year, and this number is expected to rise to EGP 105 billion by 2030. Stating the obvious, the report confirms that the problem and difficulty of driving in Cairo comes as a result of poor urban and traffic planning and management, as well as a lack of order in the country. These are all the spotlights of the problem and its consequences. This leads to a question:

“If the problem were solved, would there be a real change?” Here is the answer.

If the problem was solved:

- ❖ There would be fewer accidents and less chaos on the roads.
- ❖ People would save so much time and fuel every day.
- ❖ The air would be cleaner because cars wouldn't stay stuck for hours.
- ❖ Everyone's stress levels would go down.
- ❖ Egypt would look more modern, and people would actually enjoy moving around the city.

If the problem wasn't solved:

- ❖ Accidents will keep increasing, especially in crowded squares.
- ❖ The economy will lose more money because of wasted time and fuel.
- ❖ Pollution will rise, making the air even harder to breathe.
- ❖ People will stay angry, stressed, and tired every day.
- ❖ The country's development will stay slow because traffic affects everything.

Research

Case Study: Public Transport Problems and Their Role in Traffic Congestion in Greater Cairo

A notable 86% of respondents depend on public transport; some use it exclusively, while others combine it with alternative transportation methods.

24% of those surveyed said they spend less than EGP 50.

18% indicated their spending falls between EGP 50 and 100.

11% reported spending within the range of EGP 100 to 150.

8% claimed their spending ranges from EGP 150 to 200.

13% spend more than EGP 200.

Regarding satisfaction with public transport options, the study shows both the underground metro and inter-governorate buses received the highest satisfaction scores, at 93% each. Participants rated the micro-bus as the least satisfying choice, at 68%.

49% of public transport users rely on public buses or regular mini-buses.

72% expressed satisfaction with this form of transport.

Half of those surveyed believe service quality has improved over the last year.

40% identified traffic congestion as the main challenge for public transport users.

43% of participants use the underground metro for their transportation needs.

93% reported being satisfied with their selected method of transport.

More than half of the respondents, 54%, feel that service has progressed compared to the previous year.

Over a quarter of those surveyed, 28%, pointed out that traffic congestion is a notable challenge they face.

59% of public transport users use the tram.

72% expressed satisfaction with this mode of transportation.

Half of these respondents felt the service has improved compared to the past year.

40% mentioned traffic jams as a significant problem they encounter.

49% of public transport users depend on public buses or private mini-buses.

72% stated they are satisfied with this form of transport.

Half of those surveyed believe the service has improved compared to last year.

40% pointed out traffic congestion as a key issue affecting public transport users.

85% of respondents use private micro-buses for their transportation needs.

68% conveyed their satisfaction with this particular transport option.

41% observed no change in service quality compared to the previous year.

24% cited traffic congestion as a significant issue they face.

17% mentioned that rising costs were among the challenges encountered.

15% raised concerns about aggressive driver behavior.

60% of those surveyed reported using regular taxis for transport.

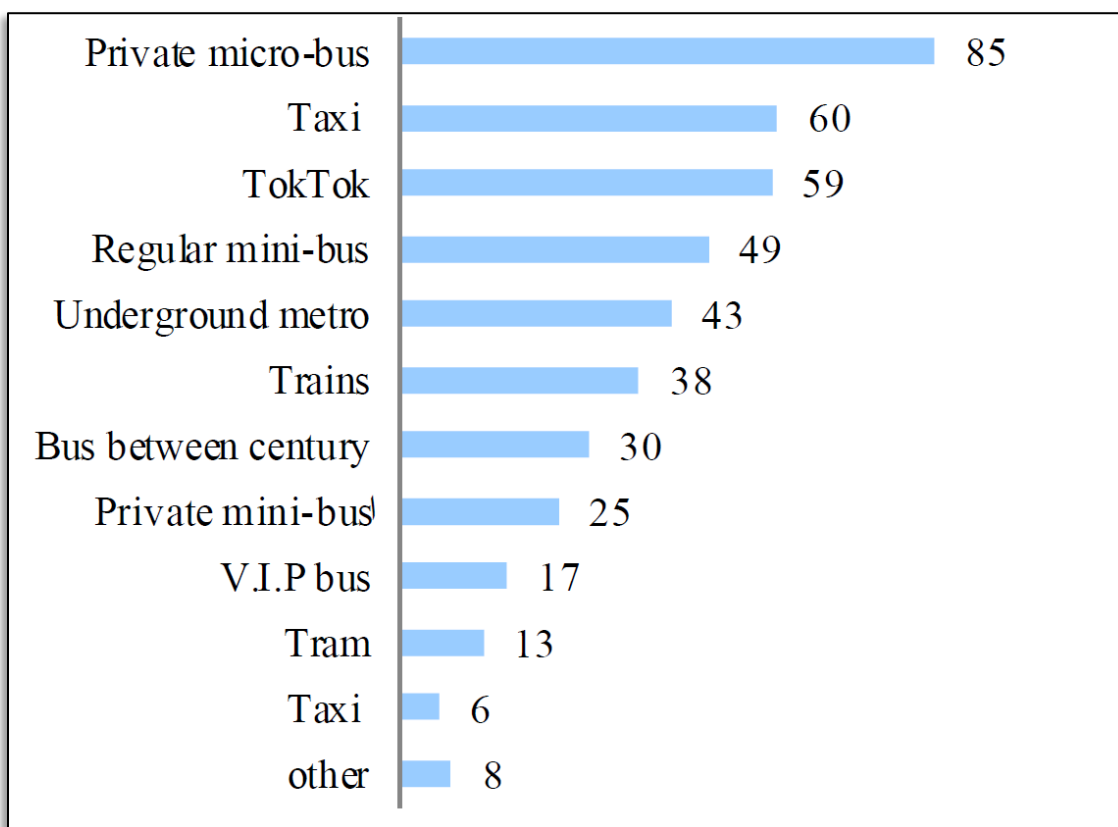
80% of respondents expressed satisfaction with this mode of transportation.

43% noted no change in service quality from the previous year.

Participants highlighted high fares and greedy drivers as significant problems.

No issues were reported concerning the Cairo Cab, which is mainly seen as an entertainment option rather than a standard means of transport.

Regarding the challenges faced by public transport users, the findings show that traffic congestion is a major problem for microbus users, public bus riders, regular mini-bus passengers, metro users, cooperative society-affiliated mini-bus riders, and train travelers.



Graph 6: the number of different types of vehicles used by people in public transport percentages.

Information and Communication Technology (ICT)

Vehicle to Infrastructure (V2I)

(Vehicle-to-Infrastructure) means communication **between a vehicle and surrounding infrastructure.**

It works using wireless technologies that can exchange the data within milliseconds, such as:

DSRC (Dedicated short-range communication)

V2X (Vehicle to everything using 5G networks)

The surrounding infrastructure, such as traffic lights, road signs, or any sensors equipped with smart sensor transmitters that constantly collect information like traffic light status and timing, speed limit, road conditions, and road signs.

The cars send information like their speed, location (GPS), brake status, and direction using OBUs (small computers with antennas that send and receive data)

V2I advantages.

- ❖ Improve road security: Warning drivers about red lights, working areas, accidents ahead, or if the car before pressed the brake.
- ❖ Less traffic congestion: Smart traffic lights adjust timing based on real-time vehicle data. By providing the other road signs and their statistics to the driver, he may leave a congested road to another free one.
- ❖ Lower fuel consumption and harmful gases emitted: by decreasing the time cars stop.

- ❖ Faster emergency response: Ambulances and police cars can control traffic light signals.

V2I disadvantages

- ❖ Data privacy issues: The cars constantly send their direction and location, and hackers could attack this data
- ❖ High cost
- ❖ Cybersecurity Risks: Hackers could potentially interfere with signals or send false data.
- ❖ Compatibility Problems: Not all cars or cities use the same V2I standards (e.g., DSRC vs 5G).

Vehicle to Vehicle (V2V)

V2V is basically giving cars the ability to gossip about everything happening on the road in real time. The ScienceDirect overview describes it as a moving network where every car becomes both a sensor and a broadcaster. Instead of waiting for a driver to react or for a traffic light to detect something, cars just tell each other what's going on: "I'm braking hard," "There's a crash 200 meters ahead," "I'm changing lanes," "Road is slippery," and so on.

The tech underneath is pretty solid. DSRC keeps delays extremely low, so messages arrive fast enough to actually matter at highway speeds. C-V2X pushes it further by letting cars talk directly *and* use 4G/5G when needed. The onboard unit in every vehicle mixes GPS, speed, sensor data, and even AI-based predictions before sending it out. The point is to create a shared awareness — not one car thinking alone, but a swarm of vehicles constantly updating each other.

The page stresses that safety is the strongest benefit. Most dangerous accidents happen because humans simply don't see the threat early

enough — blocked views, blind curves, fog, bad lighting. V2V kills that problem by making cars “see” around corners and beyond their cameras. Traffic becomes smoother too, because cars don’t behave blindly; they match speed, react earlier, and avoid sudden slowdowns that cause congestion.

But of course, nothing comes free. Cars broadcasting their movement can expose privacy issues, and a malicious sender could theoretically fake data unless the system uses strong authentication. Another challenge is how chaotic the network is — vehicles enter and leave each other’s range constantly, so the communication protocols must be light, fast, and self-adjusting.

If anything, the overview makes it clear that V2V is less of a fancy feature and more of a backbone for future transport systems. Autonomous cars without V2V are basically half-blind. Smart traffic without V2V is mostly guesswork. When cars can share real-time data directly, you get a level of coordination and predictive ability that no traditional traffic system can match.

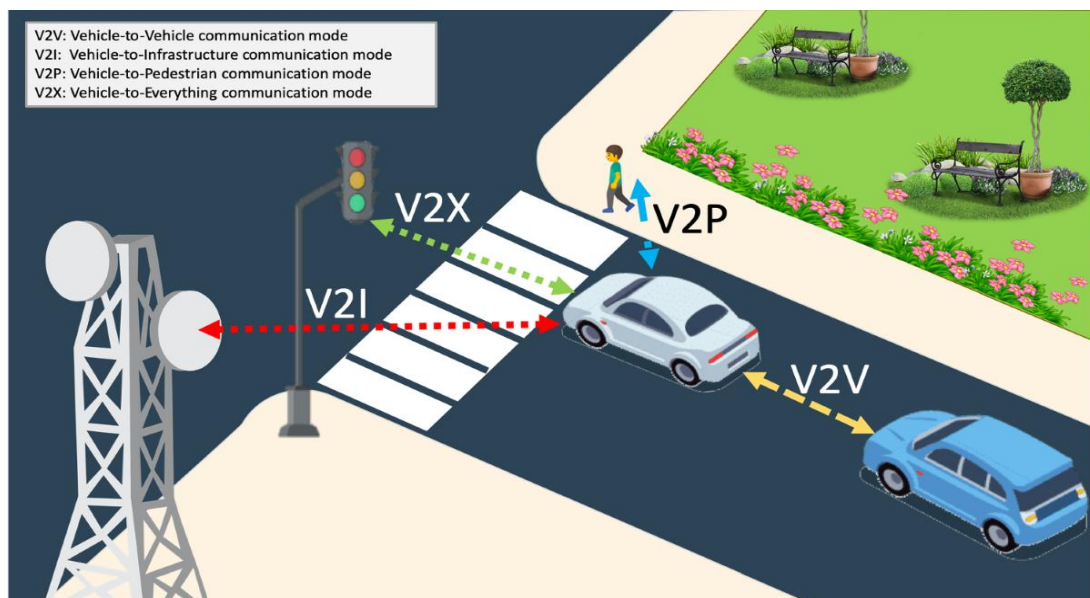


Figure 5: The different types of communication systems.

Electronic Components

ESP32-CAM

Description

The ESP32-CAM is a small microcontroller that comes with a built-in camera, and even though it looks simple from the outside, it actually handles a lot more work than you would expect from something this size. It has Wi-Fi, a dual-core processor, and enough memory to process images quickly. The camera attached to it is an OV2640 module, which captures fairly clear pictures that are more than enough for detection tasks. This board is used in many small automation projects because it does not need any external equipment to start working — everything is already integrated inside it. Because of its small size, it doesn't take much space when installed in tight areas, and it still provides a stable wireless connection. It can take photos on demand, stream video, and communicate with computers or other devices without the need for separate routers or cables. This makes it extremely practical when building prototypes where space is limited and where everything needs to run on low energy. Overall, the ESP32-CAM works like a compact surveillance unit that can see, send information, and react immediately.



Figure 6: ESP32-CAM.

Usage in the System

In our prototype, the ESP32-CAM is responsible for taking pictures of the intersection so the system can understand the traffic situation. Instead of placing a camera for every lane, the ESP32-CAM is set on a rotating base and captures four pictures — one for each road — during every cycle. These images go directly to the computer running the AI model, which

counts how many cars are present at each moment. The timing of the green lights depends almost entirely on the images taken by this board, so its role is essential. Without it, the system would not know which road is crowded and which one is empty, and the traffic lights would work blindly like normal ones. The ESP32-CAM gives the project “vision,” allowing the system to respond to real data instead of fixed times. That is the main idea of smart traffic: reacting instantly to what is actually happening on the road, not what the system “thinks” might happen.

Ultra Sonic Sensor

Description

The ultrasonic sensor is one of those components that looks very simple — two small metal circles on a small board — but the idea behind it is actually smart. One circle sends out a sound wave that we can’t hear, and the other one listens for the echo when the sound bounces back from something in front of it. The sensor measures how long that takes and turns it into a distance. It doesn’t depend on light at all, so it works in the dark exactly the same as in daylight. This makes it useful in situations where a camera wouldn’t help much. It’s also fast, and the readings are stable enough as long as the objects are not too far. You find this type of sensor everywhere: automatic doors, parking sensors, simple collision-avoidance circuits, and small school projects.



Figure 7: ultrasonic sensor.

Usage in the System

In our project, we use the ultrasonic sensor to check the distance between cars. One of the main causes of traffic issues in real life is when a single car brakes suddenly and every other driver behind reacts too late. That

delay spreads backward and the whole lane slows down even if nothing happened. The ultrasonic sensor helps us model that problem. By constantly measuring how far each car is from the next, the system can understand when the distance becomes too small. This gives us a way to keep cars moving smoothly inside the prototype instead of stopping randomly. It also supports the idea that sensors inside vehicles can reduce unnecessary braking and help traffic move in a steadier way.

Servo motor & Stepper motor

Servo Motor – Description

The servo motor is a small motor that doesn't spin freely. It only moves to certain angles, almost like it's "locked" into controlled positions. Inside it there's a bunch of gears and a tiny circuit that keeps checking where the motor is actually pointing so it doesn't go too far. It's mainly used in projects where you don't want a full rotation — just a certain amount of movement — like opening a flap, steering something, or pushing a small mechanism.



Figure 8: servo motor.

It's not a strong motor, but it's predictable. You tell it "go to this angle," and it just goes there and stays. That's basically the whole reason people use it: you don't have to keep correcting it or checking if it overshoot, because it usually doesn't.

Servo Motor – Usage in the System

In our prototype, the servo handles the movement of the green-corridor fences. We connected the servo to a 3D-printed piece that presses on the syringe, and the pressure from that syringe lifts the fence through the water-filled tube. It sounds simple, but if the servo wasn't controlled, the whole thing would jerk too fast or unevenly and the fence wouldn't rise

properly.

So the servo's job is basically to move slowly and accurately. When an emergency vehicle needs a path, the servo rotates just enough to push the actuator and raise the fence. That's it — nothing complicated, but it has to do that motion in a steady way or the whole mechanism stops working consistently.

Stepper Motor – Description

The stepper motor works differently. Instead of moving to “this angle” like a servo, it turns in tiny little steps. Every step is the same size, and the motor doesn't move unless you tell it to, so it ends up being very accurate even though it doesn't have any feedback system inside it.



Figure 9: stepper motor.

You usually see this type of motor in things that need repeated, precise movements: 3D printers, old scanners, CNC machines... stuff like that. It's not meant to spin fast; its strength is that it can move bit by bit and stop exactly where you expect it to stop without drifting.

Stepper Motor – Usage in the System

We use the stepper motor to rotate the ESP32-CAM. Instead of having four separate cameras, the stepper turns the camera to look at each road one by one. It rotates roughly 90 degrees each time and stops so the camera can take the picture. If it rotated too far or too little, the angle would be wrong and the AI wouldn't get a clear view of the road. The stepper basically keeps the scanning cycle organized. It moves → stops → camera takes the image → then it moves again. This simple loop is what makes the whole “smart intersection” idea work in the prototype, because the photos stay consistent instead of being random.

Prior solutions

Electric vehicles

Transportation is a major source of air pollution and the largest source of heat-trapping emissions in the United States. Vehicles that burn fossil fuels to move emit more than half of nitrogen oxides in our air, and are a major source of heat-trapping emissions in the US. A solution that was developed for this problem is to use a green source of energy in transportation, The electricity.

The design of electric cars has been developed since 1800, it was the first birth of cars run by electricity, the developments were just single projects and weren't made for commercial purposes.

Electric vehicles ran into a race with fossil-fueled vehicles in the early 19th century to see which will show more reliability, and at this time the fossil-fueled vehicles showed more reliability, as using fossil fuels as a source of energy actually showed reliability in other fields like powering factories, agriculture and other transportations like trains, so it was the direction of scientists to construct cars.

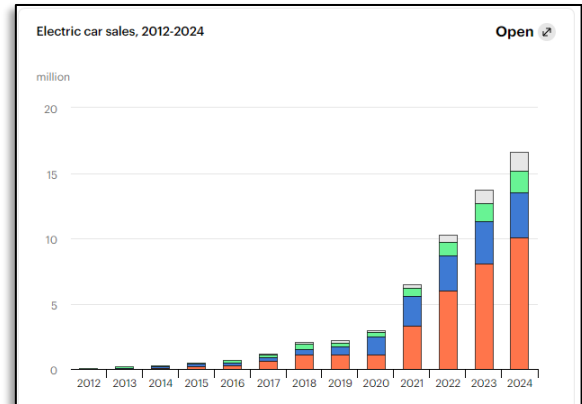
The late 1960s and early 1970s. Soaring oil prices and gasoline shortages. Created a new challenge for the United States to lower the country's dependence on Fossil fuels energy, which led to an increase the use of EVs instead of other vehicles.

for the last few decades, a lot of campaigns have been made to



Figure 10: (shows an old design for an electric car in 1901)

aware people of the consequences of air pollution and how burning fossil fuels affects our air, these campaigns worked as advertisements for electric car companies, which led to a huge increase in the usage of electric vehicles and a huge development in the designs of electric cars. And this was the biggest growth for the EVs *as shown in Graph (1)*.



Graph 7: (shows the increase in the usage of EVs)

Advantages:

- **Environmental Friendliness:** EVs produce zero emissions, reducing air pollution and greenhouse gas emissions.
- **Energy Efficiency:** EVs generally have higher energy efficiency compared to internal combustion engine (ICE) vehicles, resulting in lower fuel costs over time.
- **Lower Maintenance Costs:** EVs have fewer moving parts, leading to reduced maintenance requirements and costs.
- **Quiet Operation:** EVs are significantly quieter than ICE vehicles, providing a more peaceful driving experience.

Disadvantages:

- **Limited Range:** While EV range has improved significantly, it may still be a concern for long-distance travel or those without access to charging infrastructure.
- **Charging Time:** Charging an EV, especially for a full charge, can take longer than refueling an ICE vehicle. However, rapid charging stations are becoming more common.
- **Higher Initial Cost:** EVs often have a higher upfront cost compared to ICE vehicles, although this gap is narrowing.

- **Infrastructure Limitations:** The availability of charging stations, especially in rural areas or for long-distance travel, can be a limitation.
- **Battery Concerns:** Battery technology is constantly evolving, but there are still concerns about battery life, safety, and recycling.

Global Position System (GPS)

It is a satellite-based navigation system invented in the 1970s, and it works through a network of about 30 satellites that orbit the Earth and send signals continuously to receivers as phones. GPS is used to know the exact location of a person or a vehicle, and it is efficiently used in decreasing urban congestion problems in many ways.



Figure 11: GPS icon on devices.

Intelligent Traffic Management Systems

GPS plays an important role in intelligent traffic management systems. These systems use GPS data to monitor how vehicles move on the roads. When traffic becomes heavy on a certain street, the system can automatically adjust the timing of traffic lights to help cars move more smoothly. For example, if GPS data shows a traffic jam, the green light can stay on longer in that direction. This helps reduce waiting time, prevent long queues, and improve the overall flow of vehicles in the city.

Urban Planning and Infrastructure Design

GPS is also very useful for city planners and engineers when designing new roads and transport systems. By studying GPS data, they can see where and when congestion happens most often. This information helps them decide where to build new roads, bridges, parking areas, or public

transportation routes. Using GPS data in urban planning makes cities more organized, safer, and easier to travel in. It ensures that future infrastructure can handle the growing number of vehicles efficiently.

Reducing Environmental Impact

Finally, GPS contributes to protecting the environment by minimizing unnecessary driving and idle time. When vehicles spend less time stuck in traffic, they burn less fuel and release fewer harmful gases into the atmosphere. Therefore, GPS-based traffic solutions not only make transportation more efficient but also help reduce air pollution and energy waste, leading to cleaner and healthier urban environments.

Smart Public Transportation

GPS plays a crucial role in improving public transportation systems. Buses, trains, and other public vehicles can be tracked in real time, allowing passengers to know exactly when they will arrive. Transport agencies can also use this data to adjust schedules and routes depending on congestion levels. This not only increases efficiency but also encourages more people to use public transport instead of private cars, helping to reduce the overall number of vehicles on the road and easing urban congestion.

GPS and Ambulance Routes

The ambulance uses GPS to find the best and fastest route to an emergency. When a call for help comes, the GPS shows the quickest way to reach the location, avoiding heavy traffic and roadblocks. This helps the ambulance save time and get to the patient as soon as possible. GPS also updates the route in real-time if there are any changes on the road, ensuring the driver always takes the most efficient path. By using GPS to find routes, ambulances can respond faster and save more lives.

LOS ANGELES -ATSAC (automated traffic surveillance and control)

One of the largest and most advanced smart traffic management systems in the world, which started in 1984, has evolved into a fully AI-driven traffic control network.

The system consists of over 4500 cameras across the city to count vehicles, speed monitoring, and lane tracking.

Inductive loop sensors are installed in every lane under the road to count the waiting time and the traffic density, and traffic signal controllers that change the variety in signals according to the traffic density on each road to give priorities for the high-density traffic road.

Results

1.12%-20% travel time reduction

2.13% fuel consumption reduction

Pros

1. The largest connected traffic network in the world

The 4500 intersection operates as a unified system.

2. Highly accurate traffic analysis due to cameras and inductive sensors
3. Emergency vehicle support.

Fire trucks, ambulances, and police get priority automatically



Figure 12: ATSAC control panel.

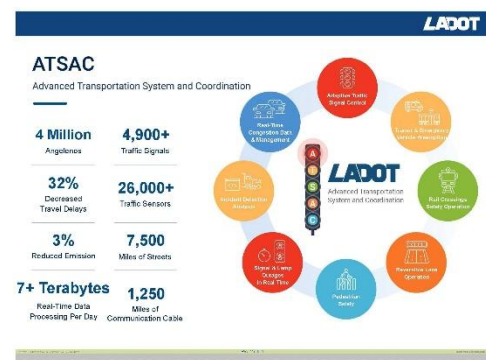


Figure 13: ATSAC system.

Cons

1. Very high cost

Millions of dollars annually, hard to implement in lower-income countries

2. Privacy concerns

Constant video monitoring raises citizen privacy issues

3. Requires strong internet connectivity

Chapter 2 | Generating and defining the solution

Solution and design requirements

Design requirements are the main standards and limitations a system or product must adhere to. These include the product's performance, user experience, safety, looks, and functionality to ensure it serves its intended purpose and meets the design requirements specified in our project:

Design requirements

- ❖ Identifying at least three measurable design requirements, including time response and system accuracy, for their prototype.
- ❖ The prototype must improve the traffic quality and safety in terms of the following (design parameters): trip time, fuel consumption, traffic comfort, accident rate, traffic flow efficiency, waiting time at intersections, emergency vehicle priority, and pollution reduction.
- ❖ Measurable change in traffic quality or safety after using the available solution.
- ❖ Testing the system response time in accordance with the system input and processing.
- ❖ Using ICT (Information & Communication Technology) is Essential. Using new trends of Artificial Intelligence (IOT and machine learning).
- ❖ The prototype must contain hardware components.
- ❖ Conducting documented calibration for sensors.
- ❖ If external sensors are used and installed on an actual road, they must provide a video showing the installation process.

Solution requirements

❖ Safety:

The traffic system must be safe by avoiding accidents with a quick response. Providing a safe space between vehicles and ensuring non-surprising actions with each vehicle controlled by the system's rules. Accident rates will be at an all-time low.

❖ Eco-friendly:

The system will apply to electric vehicles to reduce greenhouse gasses production. reducing climate change by adding sensors for air pollution detection along the road. In addition, the system's procedures for less traffic make less polluted areas

❖ Durability:

The sensors, the camera, and the server must be durable over time. The system mustn't have errors or corruption caused by a virus or other factors to avoid damage to the system, thereby enhancing the safety of the road.

❖ Efficiency:

The chosen devices for detection, organization, and even the green corridor must be able to carry out the job with zero negative effects, handling their positions, giving the highest results. With a high reaction response to actions, the reduction of air pollution, traffic congestion, and the accident rate is also achieved.

❖ Cost:

The system applies the cost reduction by making people get fewer electric cars, as it depends on the system, with some devices only applied to each vehicle. These devices connect all the vehicles with the system at a few cost instead of all buying electric vehicles at a high cost.

Selection of Solution

Our goal is to solve the traffic congestion problem, and we will do so using mechanisms and techniques we selected after extensive research and analysis of prior solutions and our subject's learning outcomes.

To make the best innovative solution, we divided the solution into main parts to solve each problem that is faced.

- **Smart traffic light**

The traffic signs will be timed according to the density of cars in each road, cameras, ultrasonic sensors and ai system will be used to count the cars and using the time of varying the traffic color will be calculated according to the cars amount, this will help in rush hours, as there will be main road all the cars are using and the another streets are free, by (time, amount) function the high density riad will have long time in green light.

- **Constant distance maintenance between each car.**

One of the reasons for the traffic congestion is the sudden stopping of the cars. If any car stops suddenly, the cars back on it will also stop, and drivers will take a few seconds to take action; these few seconds will affect all the next cars' speed, so the traffic will take more hours to get back to its normal state. This problem will be solved by maintaining a constant distance between each car; if any car stops, there will be enough distance for the following cars to take action. Also, using V2V communication technology, the following car we be alerted if the front car presses the brake.

- **Emergency road (green corridor)**

It will be used as an extra lane for emergencies, such as for ambulances and police cars. Additionally, it will be utilized during

rush hour, and if there is maintenance work on any lane, this lane will be closed using a roadblock to prevent unauthorized use.

Selection of Prototype

After researching the solution that we have chosen and our project objective, which is to make an AI-based smart traffic system.

The components of the prototype:

hardware parts:

A 120*80cm wooden plane will be used to simulate the road base, 4 roads, 3 of them are of 20 cm length, the last one is of length 60 cm representing a highway; all roads are of width 25cm, with two ways each of 10cm with an emergency road in the middle of them.

All other hardware can be represented as integrated systems:

hydraulic lifting system: moves the fences of the emergency road up and down, allowing the system to increase road width in case of no emergency and high road congestion. The system consists of:

a servo motor: to be controlled to rotate clockwise & counterclockwise using a microcontroller.

3d printed linear actuator: to convert the rotation motion of the servo into linear motion for the fence application.

2 syringes and a water pipe, which allows transferring the linear motion to where it acts with a tiny loss of energy.

wooden motion converter: consists of multiple connected wooden parts

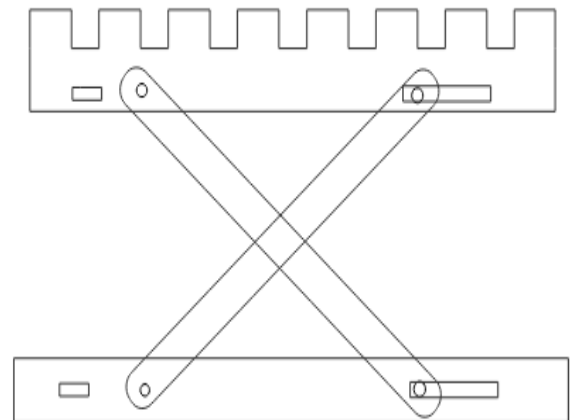


Figure 14: 2D design of the hydraulic fences lifting system.

that change the vertical motion from the syringe into a horizontal up & down motion in the fences.

Camera system: an ESP32 camera installed at the middle of the road intersection in a rotating base connected to a stepper motor, the steps of 90^0 , pointing towards each road.

Traffic lighting system: 4-2d laser cut wooden traffic of 14 cm height with 3 LEDs (green-yellow-red) for each road.

System's motherboard: an ESP32-based board with all the controlling components of all systems soldered in it, containing the following:

2 HC595 ICs: a shift register that allows saving a byte of 8 bits and open and close 8 different circuits, which is used for controlling the 12 LEDs.

UIN2003 stepper motor driver: a module that takes signals from a microcontroller and controls the stepper motor in the camera system.

High way car samples: esp8266 based cars with an ultrasonic sensor, 3v dc motor, motor driver, and 3.7 lithium batteries, used to illustrate the ideal distance maintenance between cars in high way system.

Networking system: a system dedicated to connecting all the elements of the of prototype in a mesh network, consists of a main WI-FI router and an integrated antenna in each component.



Figure 15: 2D laser cut traffic lights.

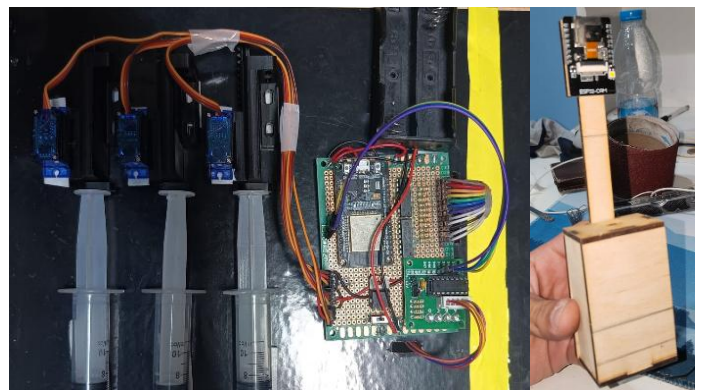


Figure 16: the motherboard of the system and the hydraulic syringes

Figure 17: the camera system

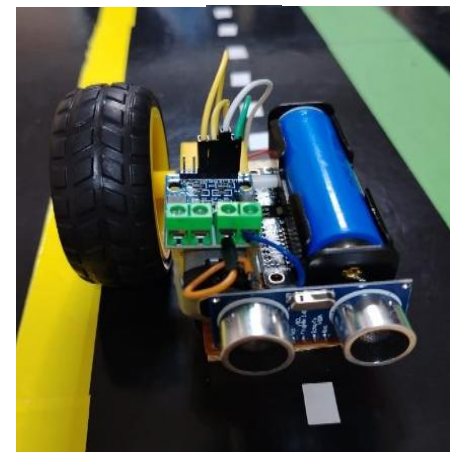


Figure 18: the car sample for the high way

Software parts:

Car detection AI models: these models take an image input and detect all the cars in it, then output the number of detected cars in the image. This model differs in our case for the prototype and the real-life application. The prototype model is trained to detect small toy cars more accurately by their colors. The real-life application uses a trained YOLO8 model to detect real-life cars on real roads and specify them as (motorcycles, cars, buses, trucks), which is utilized with more accuracy for traffic time detection.

Traffic time detection models: which use simple algorithms and calculations to determine the time needed for each road in the intersection to open according to the detected number of cars in it, with respect to other roads.

Desktop GUI application: an application that integrates all the models in it and performs the GET-POST requests for all the elements of the network, receiving the data from the camera and the sensors, and sending



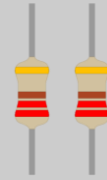
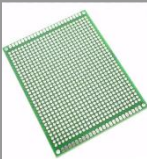
Figure 19: the whole street system

Chapter 3 | Constructing and testing a prototype

Materials and Methods

First: Table of Materials

Item	Description	Usage	Source	Quantity	Total cost	Picture
ESP 32	Small microcontroller with multiple input/output GPIO pins	Works as the main brain for the motherboard of the system, controlling all the electronic components	Electronic s shop	1	350 LE	
ESP 32 CAM	An ESP microcontroller provided with a 2mb camera sensor	Used to capture pictures of the 4 roads for the car detection and counting system	Electronic s shop	1	500 LE	
ESP8866	A tiny microcontroller with limited pins but with a WI-FI capabilities	Works in the V2V communication system to send & receive data from/to cars	Electronic s shop	3	480 Le	
li-ion 18650 batteries	Rechargeable 2000 milliamps lithium-ion batteries	Provides all parts of the project with power	Electronic s shop	4	280 Le	
Hc-595 shift register IC	An integrated circuit that allows saving a byte of 8 bits and uses it as instruction for controlling 8 bins	Used to control 12 LEDs of the traffic signals using 3 digital bins from the microcontroller	Electronic s shop	2	28 Le	

DC geared motor	Small 3-9V DC motor with a gearbox of ratio 42/1	Used to rotate the car's wheels in the highway system	Electronic s shop	3	90 Le	
LEDs	Light-emitting diode with various colors.	Represents the traffic light signals (red, green, yellow)	Electronic s shop	12	7 Le	
220Ω resistors	Electronic components that limit the current passing through a circuit	Used to adjust the current for the 12 LEDs	Electronic s shop	12	7 Le	
9*15 cm PCB breadboard	A connection board covered with copper and many holes for soldering components	The base of the motherboard, all electronic components are soldered in it.	Electronic s shop	1	15 Le	
Jumper wires	Electric wires made of highly conductive copper	Used to connect all the electronic components of the system	Electronic s shop	50	37 Le	
Sg90 micro servo motor	A motor that uses a signal from a microcontroller to control its angle of rotation	Used in the hydraulic lifting system of the green corridor fences.	Electronic s shop	3	225 Le	
Linear actuator	A gear system of a circular gear and a rack that converts rotation motion into linear motion	Connected to the servo motor to convert its rotation into linear motion	3d printed	3	120 Le	
Stepper motor	A motor that takes its signal from a microcontroller to rotate by a specific angle	Used to rotate the camera system by 90° pointing towards each road in the intersection	Electronic s shop	1	60 Le	

wooden sheet	100*120 cm - 3mm wooden sheets	Used for making the base of the prototype that the system and the cars are installed above, used also to cut some parts from the system in the laser cut machine	Carpenter	1	160 Le	
2 meter Water pipe	5 mm plastic water pipes	Used to connect the two ends of each syringe in the hydraulic system to transfer motion	Pharmacy	1	40 Le	
Syringes	10 cm syringe that carries hydraulic pressure	Used to transfer linear motion from the servo with the actuator to where it acts	pharmacy	6	40 Le	
Total cost	2509 L.E					

Second: Methods of construction

Our prototype performing methods went through several stages:

Making the prototype wooden base

The 120*80cm wooden plane was cut into 4 pieces (27.5*52.5)cm (25.5*52.5) cm (25.5*52.5)cm and (67.5*80)cm to make them easy for inserting in the laser cutter, 2 lines were cut using laser cutter with (19cm*3.3mm) on the 2* (25.52)cm and (27.5*52.5)cm for the fence, 3cm line was cut for the hydraulic lift, 4cm line was cut for LEDs wires, small hole In the center for the stepper. Using a saw, 13 pieces were cut, and using wire nails, they were fastened to the base of the plane.

Road Planning and Components

A black sticker sheet was glued onto the wooden surface to form the road background.

The road lines were added using white sticker strips.

Three traffic-light poles were designed in Illustrator and cut with the laser cutter.

Fences and Hydraulic System

Six wooden pieces were designed in Illustrator and Cut in the laser-cut to form the fence structure.

Each fence was connected to two wooden supports arranged in an X-shape and fixed with wire nails.

One of the wooden supports was connected to a water hose, which was attached on the other end to a syringe to operate the hydraulic mechanism.

Hardware system's motherboard development

Which includes installing the ESP32 microcontroller in the hollow PCB bread board, soldering the power source for the whole system, solder a switch to turn the system on and off, solder all other electronic components used to control the system including (two shift register 74hc595, 12 220-ohm resistors, 12 pin headers for connecting 12 traffic LEDs, 3 3-pin headers for the Servo motors connected to(vcc, gnd, PWM signal), the stepper motor 28BYJ-48 driver board ULN2003, and soldering a 5V voltage regulator to power the servos and the stepper motors.

Camera system installment

Software system's motherboard development

The code for the ESP32 of the mother board was made by combining some open-source codes that runs different components in the prototype in one code, then added some lines for networking that make the ESP

connect to an internet WI-FI network and run as a server with a number of Endpoints that allows controlling all project's parts through WI-FI HTTP requests.

Traffic system's camera construction

An ESP32 camera was connected to a laptop using a usb module and a camera web server example code was uploaded to it to make it connect to the WI-FI network, and adjusted the image capturing settings from the code and added an /capture endpoint that allows capturing images to any device in the network. Then a wooden box was cut in the laser cut machine to contain 2 lithium rechargeable batteries, with a hole in the bottom make the box attach to the stepper motor, then an 18cm wood was cut to attach the camera to it.

Software traffic system's AI model

A model that is trained to detect the colors of the toy cars from images was built using OpenCV library in python.

Software green time calculator model

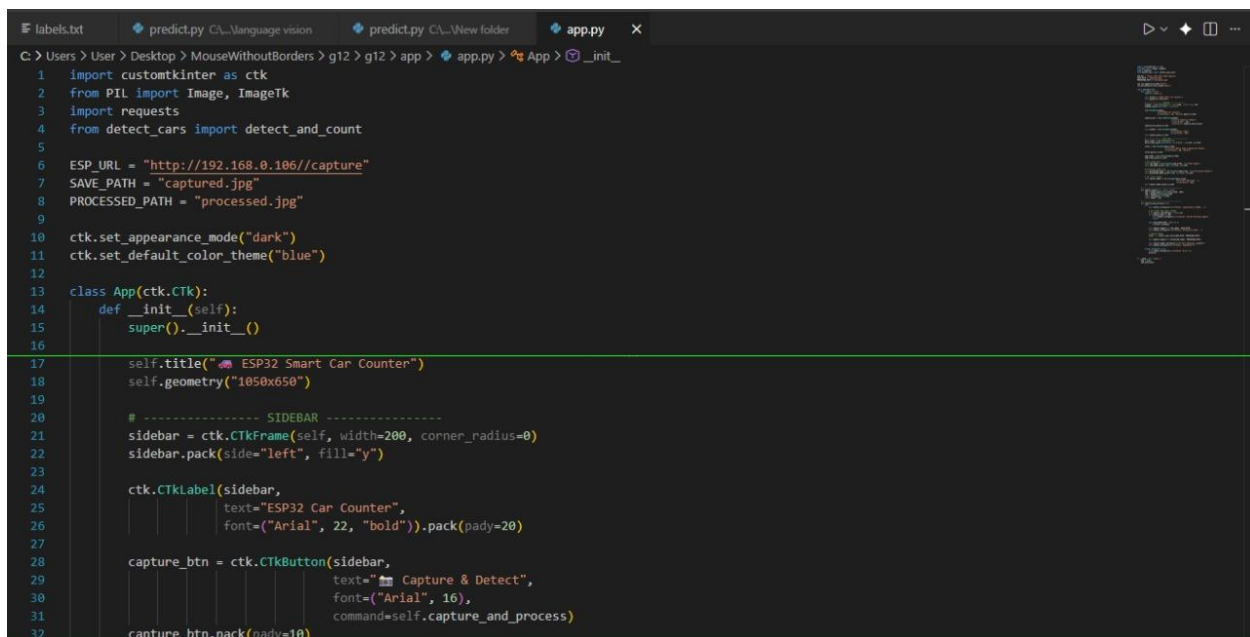
A simple python code was built to take the output from the AI car detection model and calculate the green time according to a specific equation.

Software desktop application development

A Python desktop application was made to control the whole system through a specified user interface, a Python open source library called tkinter was used to create the application's graphical user interface (GUI), and added network capabilities that makes the app connect to the mother board if it is available in the same network, then the app is able to control the mother board through the HTTP endpoints uploaded to the mother board. Then integrated all other system's models (car detection, green time calculator) into that desktop application to make decisions for the road system and apply them instantly.

Motherboard installment

After building the motherboard and testing its performance with the desktop app, it was installed into the prototype road by connecting the 12 pin headers in the board with the 12 LEDs in the traffic lights, and connecting the servos in the mother board with a linear actuator printed in the 3D printer that uses a gear and a rack to make the servo push and pull the syringe in the hydraulic lifting system, and then the stepper motor was installed in the hole in the middle of the road intersection to install the camera system above it.



```

C:\Users\User\Desktop> MouseWithoutBorders > g12 > app > app.py > App > _init_
1  import customtkinter as ctk
2  from PIL import Image, ImageTk
3  import requests
4  from detect_cars import detect_and_count
5
6  ESP_URL = "http://192.168.0.106/capture"
7  SAVE_PATH = "captured.jpg"
8  PROCESSED_PATH = "processed.jpg"
9
10 ctk.set_appearance_mode("dark")
11 ctk.set_default_color_theme("blue")
12
13 class App(ctk.CTk):
14     def __init__(self):
15         super().__init__()
16
17     self.title("ESP32 Smart Car Counter")
18     self.geometry("1050x650")
19
20     # ----- SIDEBAR -----
21     sidebar = ctk.CTkFrame(self, width=200, corner_radius=0)
22     sidebar.pack(side="left", fill="y")
23
24     ctk.CTkLabel(sidebar,
25                 text="ESP32 Car Counter",
26                 font=("Arial", 22, "bold")).pack(pady=20)
27
28     capture_btn = ctk.CTkButton(sidebar,
29                                text="Capture & Detect",
30                                font=("Arial", 16),
31                                command=self.capture_and_process)
32     capture_btn.pack(pady=10)
  
```

Figure 20: Desktop GUI app

Test Plan

After the process of construction of the prototype, multiple tests need to be applied to ensure that the chosen solution is efficient in solving congestion issues, these tests must measure the chosen design requirements for the solution to achieve:

- Image car detection model accuracy
- Traffic system HTTP requests response time
- Efficiency of increase in intersection flow rate
- V2V communication system efficiency

For testing all of these parameters, a testing plan that utilizes multiple hardware and software techniques was made for each parameter.

1) Image car detection model accuracy

To measure this parameter, the model needs to be tried in different images containing different roads with different number of cars, the car samples also needs to vary in color to detect the models capability of detecting various samples, then the accuracy can be calculated by manually counting the number of cars tested in the images and the number of cars detected by the model, and using the following equation:

$$\text{Model accuracy} = \frac{\text{number of detected cars}}{\text{numbe of tested cars}} * 100\%$$

2) Traffic system HTTP requests response time

To measure how fast does each request in the traffic system is done a windows desktop software application called **postman** can be used to make testing requests and calculate the response time of them.



POSTMAN

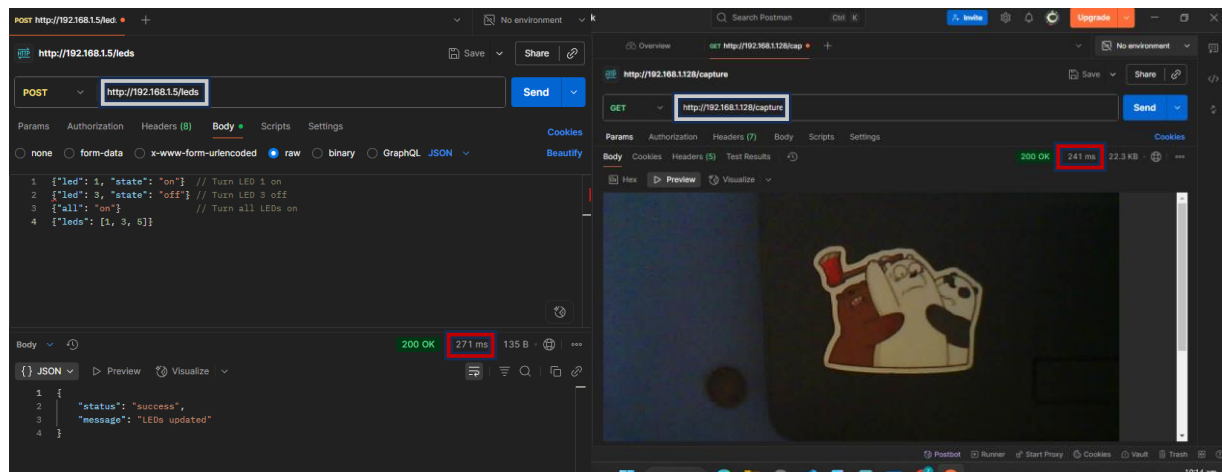


Figure 21: (illustrative image showing how to use postman to test response time of the requests)

This method is used to test all traffic system requests including (ESP32 cam image capturing, Stepper motor rotating orders, Servo motor angle control order, LEDs status updates orders).

3) Efficiency of increase in intersection flow rate

This parameter is the main concern about the traffic controlling system as it illustrates the solution main goal; the intersection flow rate represents the number of cars able to pass the intersection in a specific period of time, the goal is to maximize this quantity by controlling green time of each road according to the count of cars waiting in it with respect to the other roads.

To measure the efficiency of this system to increase the intersection flow rate, it is needed to count the number of cars passing the intersection from each road in it, which is the cars flow rate in the road (the average number of cars that passes the road in 1 second) times the green time of the road, consider that if this value is less than the actual number of cars waiting in the road, then the cars passing the roads will be just the number of cars waiting.

By calculating this for each road from the 4 intersection roads, the sum of them is the number of cars passing the intersection in one

cycle, and the intersection flow rate is the sum of the 4 roads/ Cycle time.

From the previous, it is needed to know the road flow rate to numerically determine the intersection flow rate, but for efficiency comparison between two cases, the road flow rate can be represented as a constant affecting both cases the same, and the equation for calculating the ratio between these two cases is as follows:

$$\frac{\sum_{r=1}^4 \min(\text{Road flow rate} * \text{ratio based Green Time}, \text{road cars count})}{\sum_{r=1}^4 \min(\text{Road flow rate} * \text{current systems Green Time}, \text{road cars count})}$$

4) V2V communication system efficiency

The Vehicle-to-vehicle communication system needs to be tested according to the following parameters:

- Continuous data sending speed

This parameter can be tested by calculating the rate of updating values received from each car by testing the average GET request response time of the data and solving for the updating time rate using the following equation:

$$\text{Number of requests in 1 second} = \frac{1}{\text{time of one request}}$$

- Time of deciding on an action

This is represented by the run time of the model that calculates the best distance between each car and decides how the cars should move to achieve it.

- Time of sending instructions

This is simply the average response time of the POST request to the cars.

Data Collection

Data collection is the process of collecting and measuring data, information, or other variables of interest in a uniform manner that enables the collector to test hypotheses or evaluate the project's results. Data collection is an essential stage in all types of research, analysis, and decision-making. Data gathering allows you to gain first-hand knowledge and original insights into your search problem, which we're going to use on our project.

Tool	Usage	Picture
Multimeter	Detect the current that the system uses during the testing process	
Ultrasonic sensor	Detect the distance between the cars to keep a safe distance in the V2V communication process	
Measuring tape	Used for get the dimensions while working on cutting the wooden sheets as a base for the prototype	

Green time calculations:

$$G_t = (C_t - L_t) * \text{road cars} / \text{total intersection cars}$$

Where C_t is the total Cycle time, and L_t is the lost time or the yellow time.

Trial 1	Car count	Detected cars	Detection accuracy	Green time
Road 1	5	5	100%	18.2 sec
Road 2	4	4	100%	14.5 sec
Road 3	1	1	100%	3.6 sec
Road 4	1	1	100%	3.6 sec

Trial 2	Car count	Detected cars	Detection accuracy	Green time
Road 1	4	4	100%	11.4 sec
Road 2	2	2	100%	5.7 sec
Road 3	5	5	100%	14.2 sec
Road 4	4	3	75%	8.5 sec

Trial 3	Car count	Detected cars	Detection accuracy	Green time
Road 1	3	2	66.67%	8.6 sec
Road 2	4	4	100%	11.4 sec
Road 3	4	4	100%	11.4 sec
Road 4	3	3	100%	8.6 sec

Trial 4	Car count	Detected cars	Detection accuracy	Green time
Road 1	4	3	75%	10.9 sec
Road 2	3	3	100%	10.9 sec
Road 3	0	0	100%	0 sec
Road 4	5	5	100%	18.8 sec

Average detection accuracy = (total detected cars/ total tested cars) * 100

$$= (49/52) * 100 = 94.2\%$$

Efficiency of enhancing intersection flow rate calculations for each trial:

$$\frac{\sum_{r=1}^{r4} \min(\text{Road flow rate} * \text{ratio based Green Time}, \text{road cars count})}{\sum_{r=1}^{r4} \min(\text{Road flow rate} * \text{current systems Green Time}, \text{road cars count})}$$

For calculating the actual intersection flow rate, it is needed to determine the actual road flow rate, but for efficiency calculation between two systems, as in our case, the road flow rate can be considered as a common constant between the two systems.

Trial 1	(Rf) Road flow rate (constant)	(N) Road cars count	current systems Green Time (total green time/number of roads)	Ratio based Green time	Rf * current systems G _t (number of cars able to pass the road in its green time)	Rf * ratio based G _t (number of cars able to pass the intersection in its green time)	Min (Road Rf * current systems G _t , N)	Min (Road Rf * Ratio based G _t , N)
Road 1	0.333	5	10 sec	18.2 sec	3.33	6.06	5	3.33
Road 2	0.333	4	10 sec	14.5 sec	3.33	4.82	4	3.33
Road 3	0.333	1	10 sec	3.6 sec	3.33	1.19	1	1
Road 4	0.333	1	10 sec	3.6 sec	3.33	1.19	1	1
$\sum_{r=1}^{r4}$	-----	-----	-----	-----	-----	-----	11	8.6

System's
Efficiency

$$(11/8.6) * 100 \% = 128\%$$

Response time calculations.

for the calculations of the system response time, a desktop Windows application called **Postman** was used to test various APIs and endpoints of the system, providing accurate response time for each request. The results of the response time of each HTTP request involved in the system are shown in the following table:

Request	Request Type	Data transferred	Trial 1 R.T	Trial 1 R.T	Trial 1 R.T	Trial 1 R.T	AVG Response time
ESP32 cam image capturing	GET	Image	242 ms	299 ms	277 ms	238 ms	264 ms
Stepper motor rotating order	POST	JSON raw input	1.18 sec	1.31 sec	0.98 sec	1.12 sec	1.15 sec
Servo motor angle control order	POST	JSON raw input	211 ms	283 ms	97 ms	120 ms	177.8 ms
LEDs status updates order	POST	JSON raw input	207 ms	269 ms	274 ms	271 ms	255.3 ms

AI models run-time calculations:

This was simply obtained by observing the execution time of the models from the terminal for many trials and calculating the average of them.

Model	Trial 1 R.T	Trial 2 R.T	Trial 3 R.T	Trial 4 R.T	AVG Run time
Prototype Image car detection model (toy cars)	0.88 sec	1.2 sec	0.73 sec	0.51 sec	0.83 sec
Real life YOLO8 Image car detection model (real cars)	3.28 sec	2.55 sec	1.93 sec	3.03 sec	2.70 sec
Green time calculation model	130 ms	88 ms	51 ms	62 ms	82.6 ms

Average System's Cycle Time calculation:

A complete Cycle in the system includes 4 stepper motor rotating orders, 4 image capturing orders, 4 image car detection processes, 1 Green time calculations, and 1 LEDs status updating order, by combining the Average time required of all of these steps times the number of repetitions of each step in a cycle, the sum is the average System Cycle time:

$$1.15 * 4 + 0.264 * 4 + 0.83 * 4 + 0.082 * 1 + 0.255 * 1 = \mathbf{9.313 \text{ sec}}$$

Chapter 4 | Evaluation, Reflection, Recommendations

Analysis and Discussion

The solution “**Solving Traffic Congestion Using AI-based traffic lights management and V2V Communication**” works to solve two main challenges, which are urban congestion and dealing with population growth. These challenges are causing the main problem, which is traffic congestion in Greater Cairo intersections. As the Greater Cairo Region (GCR) is known to be represent about 19% of Egypt's population, as it had about 16.1 million inhabitants in 2018, and this number has grown in the past few years.

After long research, a solution has been reached, which is an AI-based traffic system that works with three main programs. A traffic light that works with the ESP32-CAM, with an AI model to count the number of cars to detect the priority of the busiest road, and give the exact time for each road to be open to save people's time. This is decided by the AI model. In addition to a green corridor for emergencies that works with a hydraulic lift system, opening when a lane is busy, or there is an accident, or even for an ambulance to move faster alone in that extra lane. Not forgetting to mention the V2V communication to keep a safe distance between the cars and avoiding the late response time of humans.

After analyzing how the idea works, some design requirements were tested to make sure whether they were achieved or not. The traffic light system was tested to determine the time that the intersection takes in one cycle. Making four trials, changing the number of vehicles each time to check the system detection accuracy, the result was that the average was calculated, and the average detection accuracy equals 94.2% this was estimated by the law:

$$\text{Average detection accuracy} = (\text{total detected cars} / \text{total tested cars}) * 100 = (49/52) * 100 = 94.2\%$$

After getting the system accuracy as shown in graph X, the first trial got 100% accuracy, the others had the problem of that the AI model couldn't detect some colours like black, so it miscounted them. However, by analyzing the first trial, which was the most accurate. The intersection flow rate was calculated by getting the time that each car passes the intersection, and comparing the normal system green light time to the AI-based system, as shown in graph Y. The difference in allocation of time for all four roads in the intersection was based on the number of cars, resulting in less time wasted by people. This comparison led to the system efficiency, which was calculated by the traffic flow rate, according to this equation:

$$\frac{\sum_{r=1}^4 \min(\text{Road flow rate} * \text{ratio based Green Time}, \text{road cars count})}{\sum_{r=1}^4 \min(\text{Road flow rate} * \text{current systems Green Time}, \text{road cars count})}$$

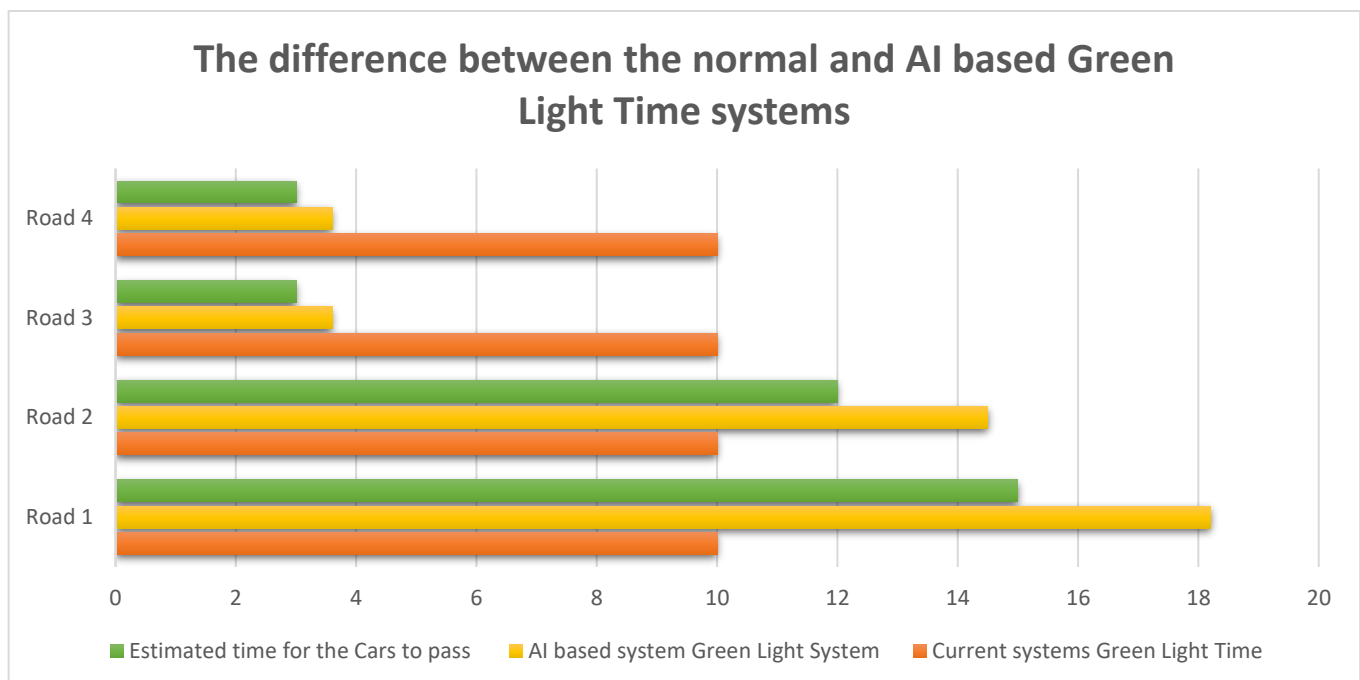
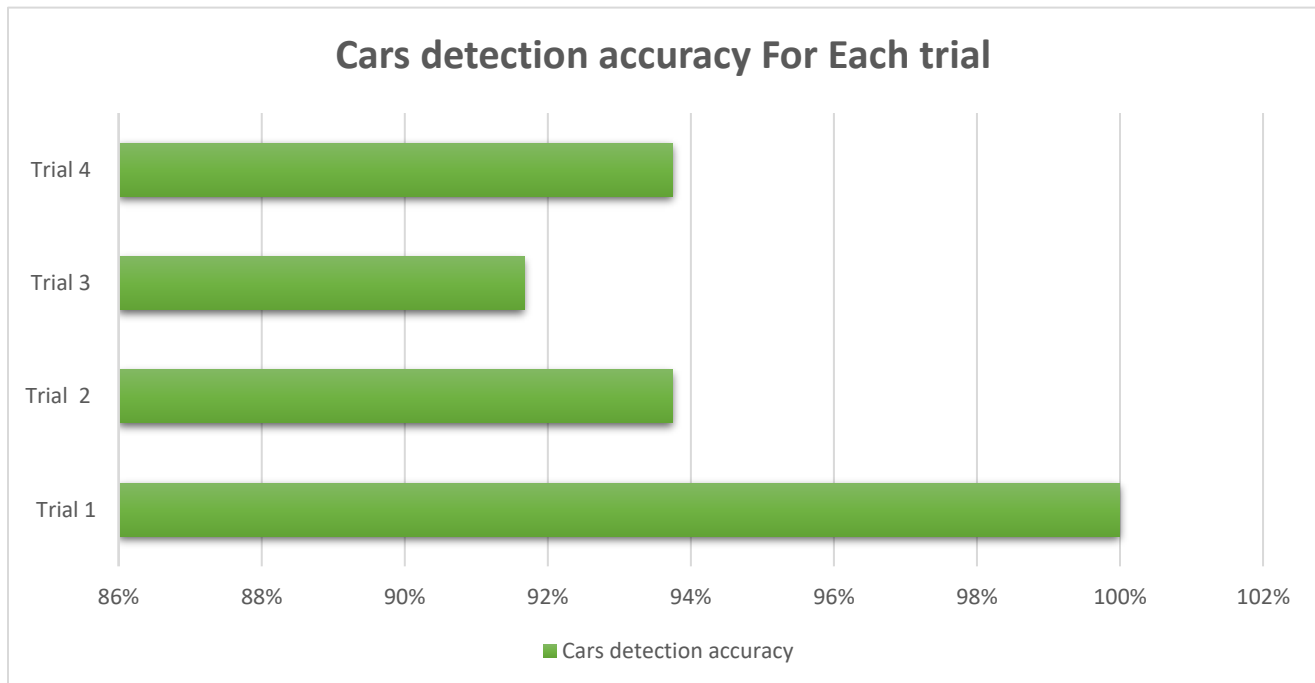
With that, the system's efficiency is 128% which is higher than the normal system with 28% in this case, as the traffic flow rate in this condition is 0.333 for each road. These results illustrate the time that is saved by the system as each road gets the exact time to be free of cars. As the first road had 5 cars, it needed 18.2 seconds to pass all the cars through the intersection, but in the normal state, the green light would take 10 seconds for each road, so the difference is shown in how to organize the priority and calculate the Green Light Time. This equation was used:

$$G_t = (C_t - L_t) * \text{road cars} / \text{total intersection cars}$$

Where C_t is the total Cycle time, and L_t is the lost time or the yellow time.

To continue, the third parameter calculated was the system time response. The time of the system working for each cycle was divided into two parts: the response time of the devices and the AI model runtime to receive data, analyze it, and send orders. For the calculations of the system response time, a desktop Windows application called **Postman** was used to test various APIs and endpoints of the system, providing accurate response time for each request. To calculate the total time that the cycle needs to

empty four roads with no errors. The detected response time of image capturing, stepper motor rotating, the LEDs status updating, and the green time calculation in the previous part. Adding the run time of the AI model to receive data from the ESP32-CAM and the counting process, and the order sending to the lights. The total time response is calculated to be 9.313 seconds.



Recommendations

"Nothing is perfect by humans." By the time new developments and technologies emerge, they often affect existing ideas or projects. This is why it's essential in scientific research to write recommendations for others or researchers who will work in the same field or on the same project, thereby avoiding mistakes.

Recommendations for real-life application:

1. Using a wide lens camera such as for the road so more vehicles can be detected, and the intersection can move faster.
2. Applying the V2V for more applications like: keeping safe distance, sending directions signals early, warnings about sudden breaks.
3. Connecting the system to signal towers along the road so all the cars get the updates faster instead of the router that works as the receptor and server at the same time
4. Using Yolo 8 AI model for a better cars detection accuracy leaving no errors.
5. Constructing a security system for the project so it gets the access to the motherboard

Recommendations for the prototype:

1. Cutting the stands for the prototype with more accuracy for more balance.
2. Apply the V2V communication system in the used cars instead of the motors and sensors only.
3. Using a smaller sample that looks like a real car for a better simulation process specifying the kinds of vehicles like minibuses and tuk-tuks so the AI model can detect the different types and count them.

During the creation of the project through the fourth semester, different challenges were faced that can affect humans positively, making a team stronger and learning from the new, invaluable skills, from these great skills:

1. Understanding the importance of solving the traffic congestion problem in Egypt and how it affects the humans future and daily life.
2. Dealing with and creating AI model, using programming languages such as python, and Arduino. Working on visual studio code.
3. For using the laser cutter, Making 2D design on Coral draw app. Enhancing the ability of creating more parts that are effective in the prototype. In addition to 3D modeling working on Blender.

Learning Outcomes

Learning outcome	Connection with the project
PH.3.01	In our prototype, we use WI-FI, which is electromagnetic radio waves to send and transfer data between the communication system components, and this type of wave is characterized by its high stability and its medium range (30m-100m), which will be suitable for our prototype.
ES.3.02	Our communication system depends on detecting cars and counting them, like the main function of the GPS in detecting the position of objects on the earth.
BI.3.01	There are great similarities between the neurons structure and our communication system, neurons divides into 3 categories; sensory, integration and motor neuron, sensory neurons that sensing the data and it travels to the integration neurons to integrate the data then the motor neuron takes the action, in our prototype the sensors like the ultrasonic sensors sense the distance and the ai mode integrate the data and send it to the motors to take specific action.

BI.3.05	reflexes in muscles are responsible for any emergency that needs a speedy response and take action, the same thing in our prototype, if any danger occurs that affects traffic, the green corridor opens.
BI.3.06	Our communication system acts as the structural backbone and keeps the traffic ecosystem stable and coordinated, like the skeletal system that provides support for the structure and holds everything together.
CH.3.01	The AI model works in the same steps of the scientific method. As it learns to work, following the observed patterns in data and develops predictive “laws.” For instance, it might deduce that traffic congestion increases
PH.3.04	The IoT system mirrors this exactly: sensors (transmitters), wireless or fiber channels (medium), and central AI servers (receivers). This structure ensures real-time, error-free communication between devices. In addition to using antennas that allow V2V communication, enhancing the data range.
PH.3.05	The system samples environmental and traffic signals (vehicle flow, speed) at defined rates. Choosing the correct sampling rate ensures that the digital data accurately reflects real-world changes without delay or data loss — crucial for real-time AI analysis and adaptive control.
MA.3.01	Using an implicit function allows your system to predict pollution levels for given traffic conditions without explicitly solving the complex formula for y, aiding real-time rerouting.
MA.3.03	Critical points in traffic density or pollution levels highlight danger zones where intervention is needed. Your system can prioritize rerouting or alert authorities when these points are reached to prevent accidents or severe congestion.

Citation in APA^{7th}

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